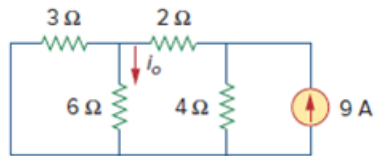


Problem

Use linearity to determine i_o in the circuit of Fig. 4.72.

Figure. 4.72:



Step-by-step solution

Step 1 of 5

Refer to figure 4.72 in the textbook.

Assume that the value of current i_o as 1 A.

Redraw the circuit as shown in figure 1.

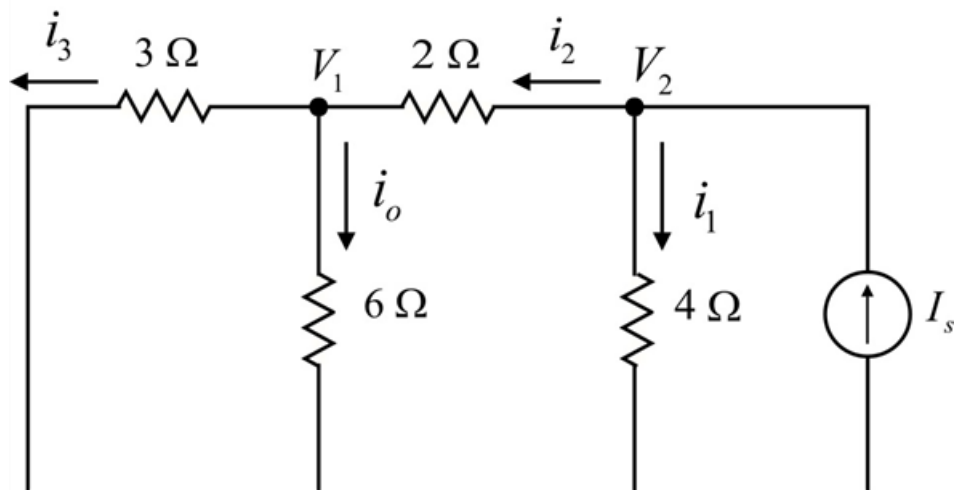


Figure 1

Step 2 of 5

Calculate node voltage V_1 .

Apply Ohm's law.

$$V_1 = 6i_o$$

Substitute 1 for i_o .

$$\begin{aligned} V_1 &= 6(1) \\ &= 6 \text{ V} \end{aligned}$$

Calculate current i_3 .

Apply Ohm's law.

$$i_3 = \frac{V_1}{3}$$

Substitute 6 for V_1 .

$$\begin{aligned} i_3 &= \frac{6}{3} \\ &= 2 \text{ A} \end{aligned}$$

Step 3 of 5

Apply Kirchhoff's current law at node V_1 .

$$i_2 = i_3 + i_o$$

Substitute 2 for i_3 , and 1 for i_o .

$$\begin{aligned} i_2 &= 2 + 1 \\ &= 3 \text{ A} \end{aligned}$$

Step 4 of 5

Calculate voltage V_2 .

Apply Kirchhoff's voltage law.

$$V_2 = 2i_2 + V_1$$

Substitute 3 for i_2 , and 6 for V_1

$$\begin{aligned} V_2 &= 2(3) + 6 \\ &= 12 \text{ V} \end{aligned}$$

Calculate current i_1 .

Apply Ohm's law.

$$i_1 = \frac{V_2}{4}$$

Substitute 12 for V_2 .

$$\begin{aligned} i_1 &= \frac{12}{4} \\ &= 3 \text{ A} \end{aligned}$$

Step 5 of 5

Apply Kirchhoff's current law at node V_2 .

$$I_s = i_2 + i_1$$

Substitute 3 for i_2 and i_1 .

$$I_s = 3 + 3$$

$$I_s = 6 \text{ A}$$

Therefore, the assumption of i_o as 1 A current is due to a current source of 6 A.

But the actual current source is of 9 A.

Apply Linear property.

$$i_o = (1) \left(\frac{9}{6} \right)$$
$$= 1.5 \text{ A}$$

Therefore, current i_o is 1.5 A.